

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the simulation of human intelligence by machines. It encompasses machine learning

Types of Machine Learning

Supervised Learning

Models trained on labeled data. The algorithm learns a mapping from inputs to outputs. Examples: classification, regression.

Unsupervised Learning

Models find patterns in unlabeled data. No target variable is provided. Examples: clustering (customer segmentation), association.

Reinforcement Learning

An agent learns by interacting with an environment and receiving rewards or penalties. Examples: game playing, robotics.

Key Concepts in Deep Learning

Overfitting occurs when a model learns noise in training data rather than the actual pattern. It results in high

A neural network consists of layers: input layer, hidden layers, and output layer. Each neuron applies a weight

| Concept | Description |

| Epoch | One complete pass through the training data |

| Batch Size | Number of samples processed before updating weights |

| Learning Rate | Step size for weight updates during optimization |

| Loss Function | Measures how well the model's predictions match actual values |